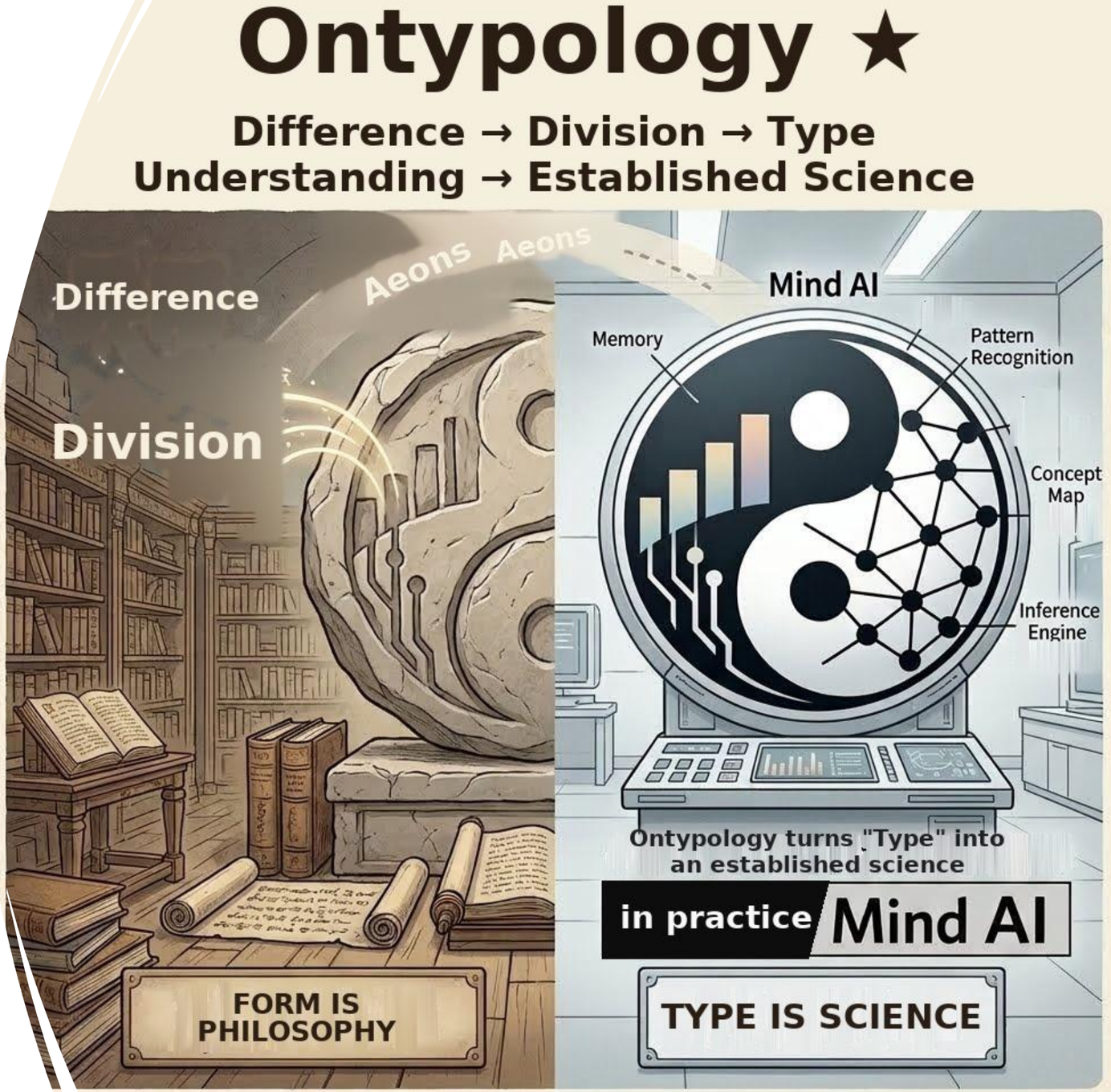
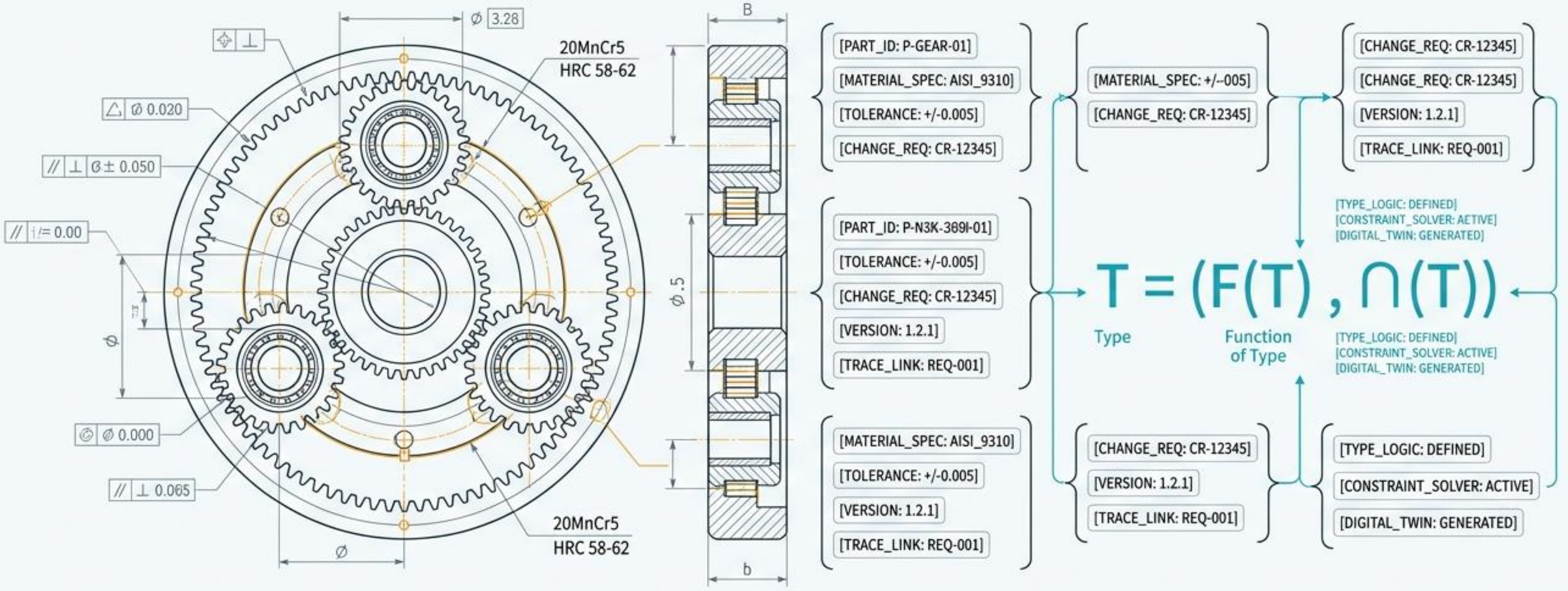
Ontypology

- From difference to division, from division to type, from type to understanding – the brain has walked this path for aeons.
- A century ago, logic was philosophy; once seen clearly, it became an established science.
- Today we explore "Type" (a brain concept) as philosophy; Ontypology makes "Type" clear, turns it into an established science, and puts it into practice as Mind AI.



the base of the AI mechanical design & PLM brain

from messy text to a reasonably inferable “product replica”: in spec, part, change and history, every number can state its origin.



[core philosophy: Type first, numbers after
(First Type, then Number)

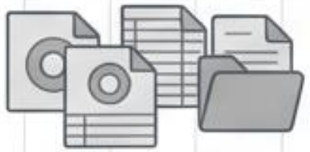
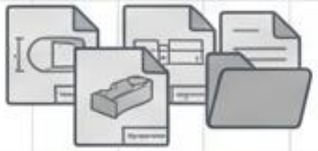
[base platform] Mind AI × LLM
(Intelligence x Language)

[final promise] output
(100% Traceability & Accuracy)

Convention one: Type in English, never Form.

Type-Driven Design

Convention two: use the theory as a tool, the platform as a base.

engineering language		Ontypology language
	specification, requirement clauses	**SpecType** (property [ ], property values, ∩(T))
	part drawing, material list	**PartType** (Unit Type, BOM set { })
	assembly drawing, BOM	**AssemblyType** (composition, Spatial Container ( ))
	inspection report, measurement data	**InspectType** (acceptance, three-layer judgement)
	engineering change notice ECN	**ChangeType** (three levels of remove, dependency-chain impact)
	revision history, configuration management	**HistoryType** (identity anchor id(C), Time Type evolution)

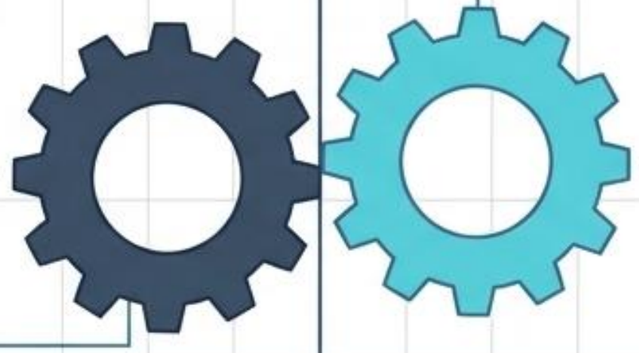
MIND AI

end the four symptoms of “inaccurate output” and the twin-engine architecture

twin-engine architecture (The Engine)

LLM (language understanding and generation)

– handles language: turns the engineer’s dialogue into SpecType and reports.



Mind Model (unified knowledge and memory)

- – handles intelligence: Types (PartType, moves library, long-term criteria, deductive reasoning, task decomposition, learning.

diagnostic matrix

Symptom 1: the same part does not match across documents

prescription: establish the PartType, anchor identity with id(C).

Symptom 2: the spec is written but cannot be accepted

prescription: build ∩(T) acceptance into the SpecType.

Symptom 3: documents left unrevised after a change

prescription: AssemblyType BOM + remove, level-3 impact analysis.

Symptom 4: AI output looks reasonable but nobody dares use it

prescription: Type evidence chain, every number sourced.

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two Types: space raises the skeleton, time supplies the moves

Space Type - what the product looks like

Y axis { }

composition: BOM levels, part composition.

(e.g. L0 machine → L1 train → L2 sun gear

Z axis ( )

container: housing, module, storage boundary.

(e.g. the planetary train is enclosed in the housing; lubricant must not cross the boundary)

X axis [ ]

property: spec clause, parameters, tolerances.

(e.g. module 1.5, pressure angle 20°, accuracy ISO 1328 grade 6)

Z axis ( )

container: housing, module, storage boundary.

(e.g. the planetary train is enclosed in the housing; lubricant must not cross the boundary)

Time Type: how R&D moves

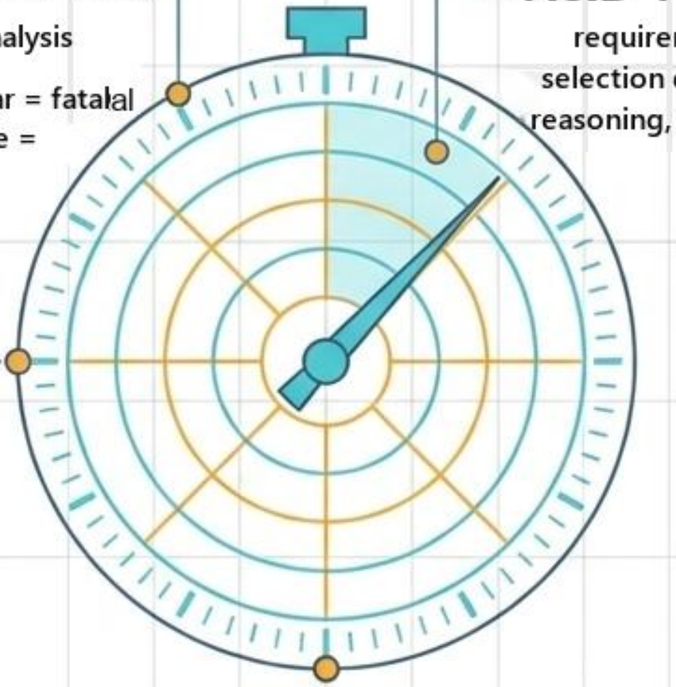
Three levels of remove:

change impact analysis

(removing sun gear = fatal
nameplate change = imperceptible)

R&D moves (14)

requirement analysis, selection design, strength reasoning, ECN refactoring.



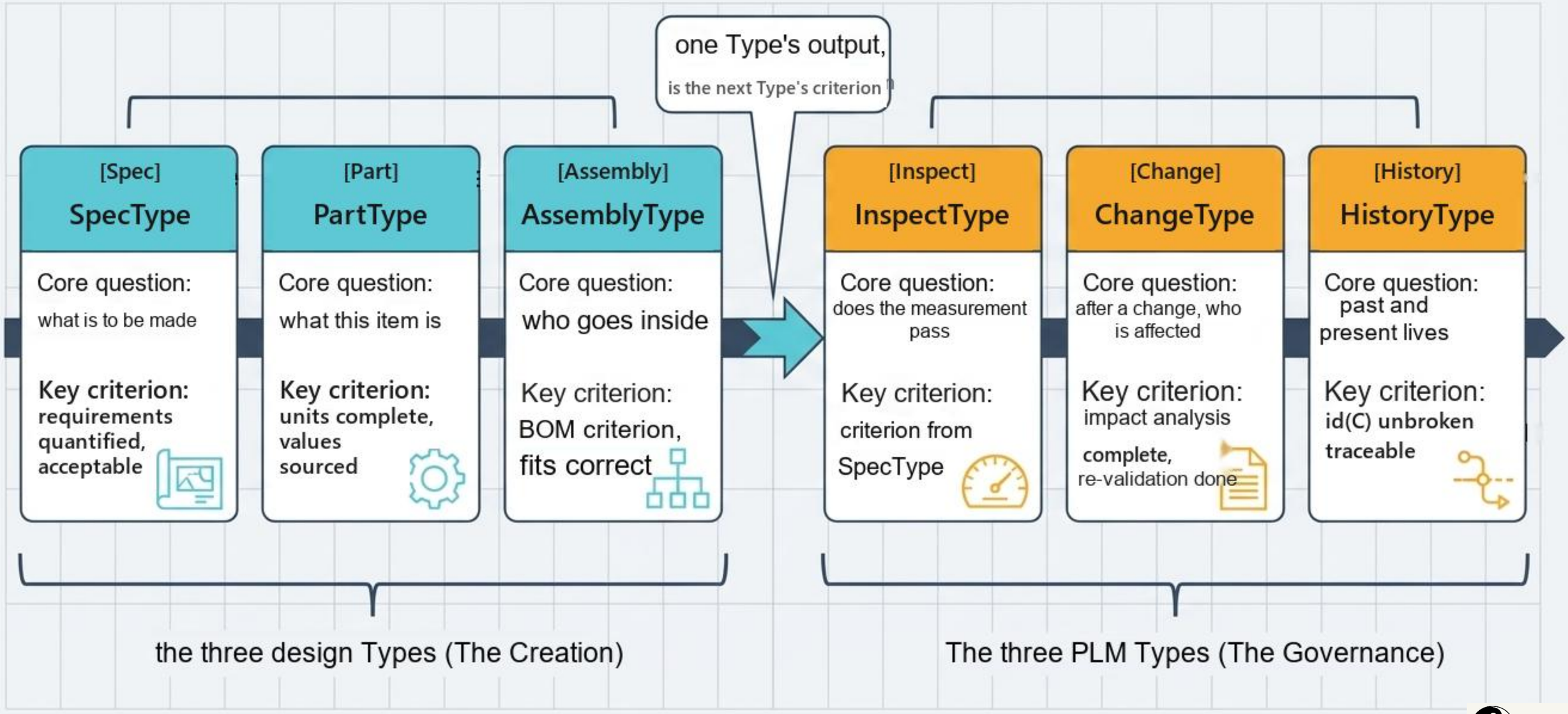
id(C)

identity anchor: configuration management

(revisions unbroken: A → B is the same machine).

Gemini Notebook

the six output Types at a glance: spec, part, assembly, inspection, change, history



the three design Types in depth: requirements into computable entities

SpecType

spec number: SPEC-PG-001

requirement clause: ratio 5:1

quantified criterion ∩(T):

$i = 5 \pm 2\%$; $T2 \geq 50\ N \cdot m$

constraint: OD ≤ 120 mm;
weight ≤ 3.5 kg

PartType
(PT-SG-018 sun gear)

material and heat treatment:
SCM415 carburised HRC58-62

key dimensions:
module 1.5; 18 teeth;
18 teeth; pitch circle Φ27 mm

tolerance: profile ISO 1328 grade 6;
Shaft Φ12 k6

AssemblyType
(AS-PGS-01 planetary train)

BOM level: L1 (under L0)

detail quantity: sun gear ×1
planet gears ×3
ring gear ×1, pins ×3

fit: planet gear bore Φ14 H7 /
pin Φ14 k6

Three levels of remove (how to handle at change time)

- ☐ Level 1 (essential): remove it and it is not itself (sun gear) → ECN required, re-validation required
- ☐ Level 2 (functional): remove it and function degrades (fan cover) → assess the degradation,
- ☐ Level 3 (non-essential): remove it, identity unchanged (nameplate) → light change

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The three PLM Types in depth: acceptance, change, unbroken revisions

InspectType (INSP-PG-001-A)

inspection item:

backlash

measurement method / instrument:
dial-indicator reversal;

resolution 0.01 mm

criterion:

SPEC-PG-001 §4:

≤ 7 arcmin (from the SpecType)

measured and judged:

0.09 mm / pass

(three values: pass / fail / concession)

ChangeType (ECN-042)

change content:

oil seal material NBR → FKM

impact analysis:

affecting PartType / AssemblyType / SpecType

(computed from remove level)

re-validation items:

150°C seal hold test

lip wear re-measurement

HistoryType

revision baseline:

Rev A → Rev B

(anchored by id(C))

bound change:

Rev B = Rev A + ECN-042

release basis:

type tests all pass

archiving:

Rev A drawing stamped void, filed for ten years

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four layers of accuracy assurance, and layer one: the ∩(T) funnel

four layers at a glance (all for accuracy; one missing, all missing)

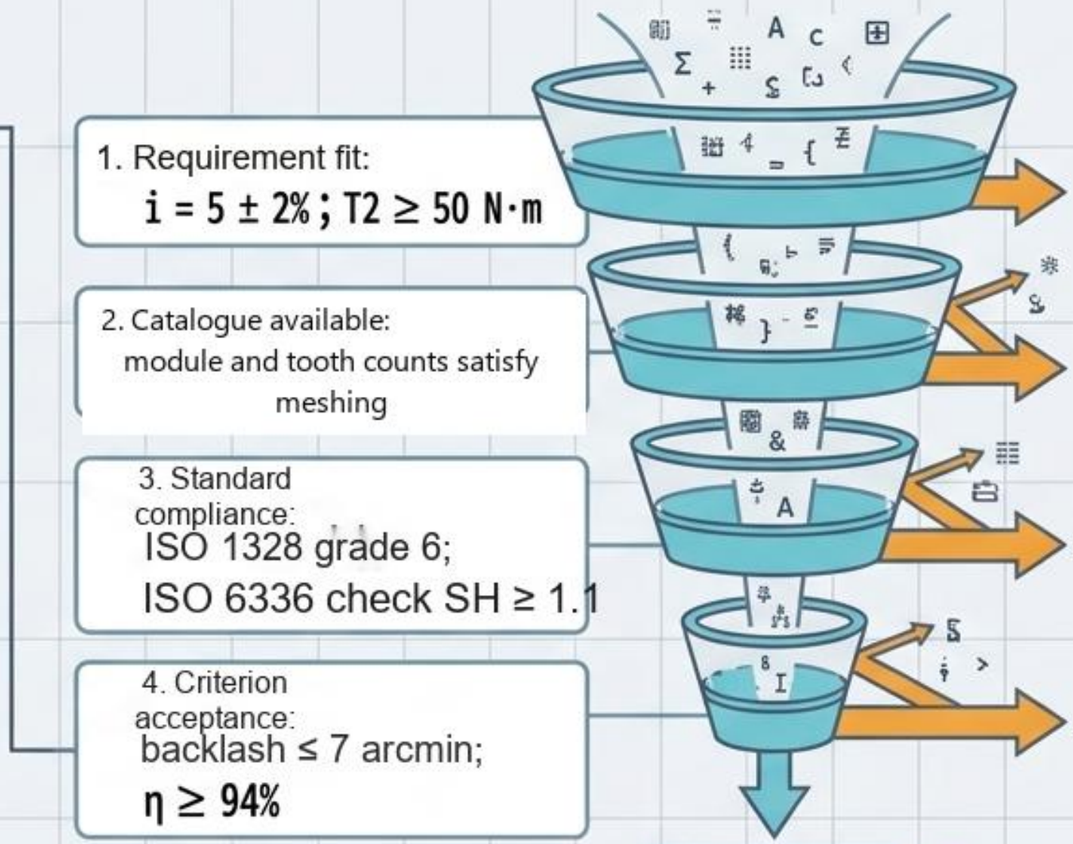
1. Eligibility: does it count as this Type
(∩(T) criterion)

2. Reasoning: how it was derived
(deductive skeleton)

3. Reality: can the numbers hold up
(engineering validation)

4. Memory: can it be found afterwards
(Type evidence)

Gate one, the four-gate ∩(T) funnel - planetary gearbox example



Gates two and three: deductive skeleton and physical engineering validation

Gate two - deductive skeleton (conclusions are derived, not copied)

derivation rule: known → formula → substitute → result (units at every step)

example step (1)

- Known: T2 = 50 N·m; i = 5; η = 0.95
- Formula: T1 = T2 / (i × η)
- Substitute: T1 = 50 N·m / (5 × 0.95)
- Result: T1 = 10.5 N·m

example step (2)

- Known: i = 1 + Zr / Zs
- Formula: Zr = (i - 1) × Zs
- Substitute: Zr = (5 - 1) × 18
- Result: Zr = 4 Zs (Zs = 18, Zr = 72)

Gate three - engineering validation (can the numbers hold)



dimension

tolerance stack-up (worst-case / statistical stack-up added to ensure the clearance passes)

strength

Strength check (ISO 6336 contact and bending, safety factor passes)

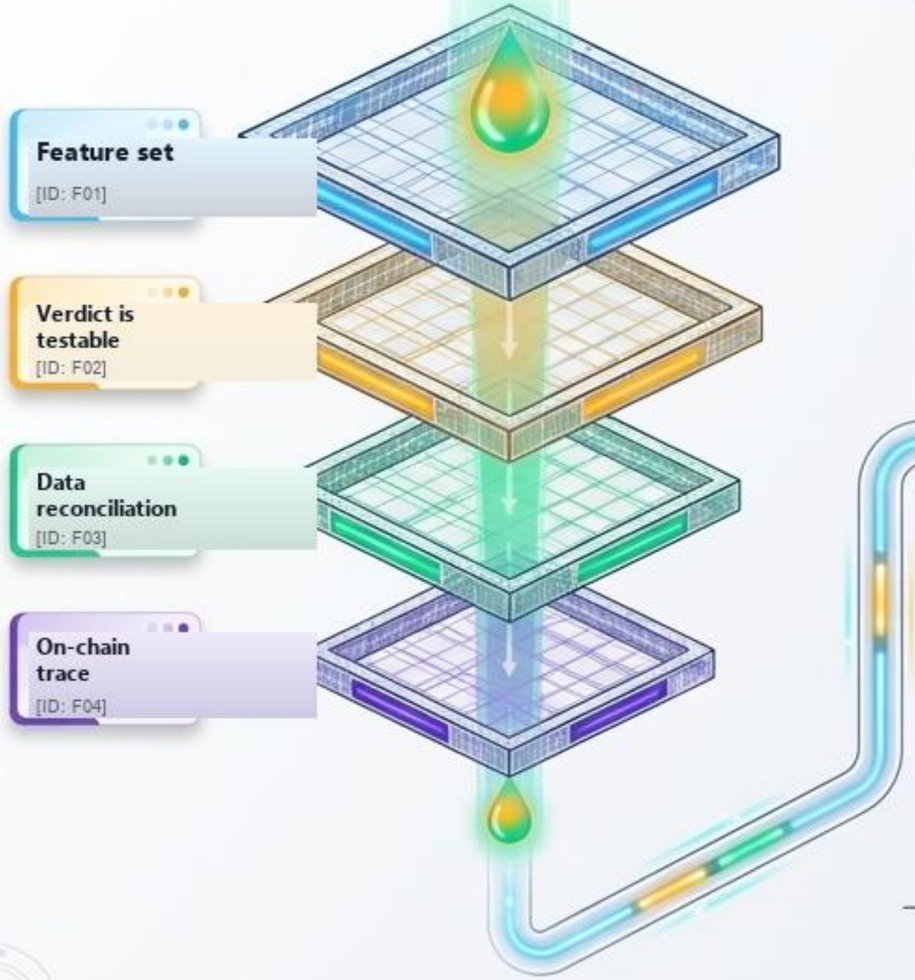
performance

performance test (efficiency, noise, temperature-rise measurement)

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Verdict funnel ∩(T)

[Filterayer]



[Horizontal Skeleton]



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Six stations of the Type evidence chain (EvidenceChain)



Three questions per station: what were the criteria? Did it pass? Where is the evidence? Any break is returned.

Space x Time Grid

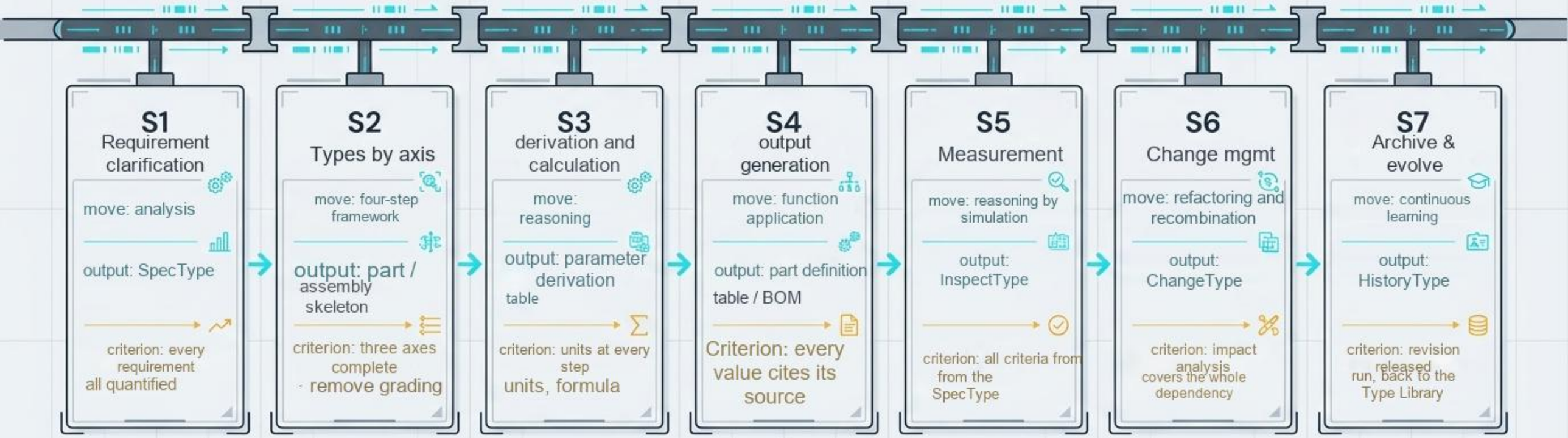
	[ ] Property layer (Rules)	{ } Composition (BOM)	( ) Container (Identity)
Design gate (AMBER)	✓ Write the features	✓ Part list	Each item tested ✓ Changing it leaves id unchanged
Evidence gate (AMBER)	✓ Full specification	✓ Full comparison	✓ Confirm the change
Decision gate (AMBER)	✓ Data archiving	✓ Risk labelling	✓ Version consistency

Only when all cells are filled does the verdict document deserve a signature. A blank cell means no verification.

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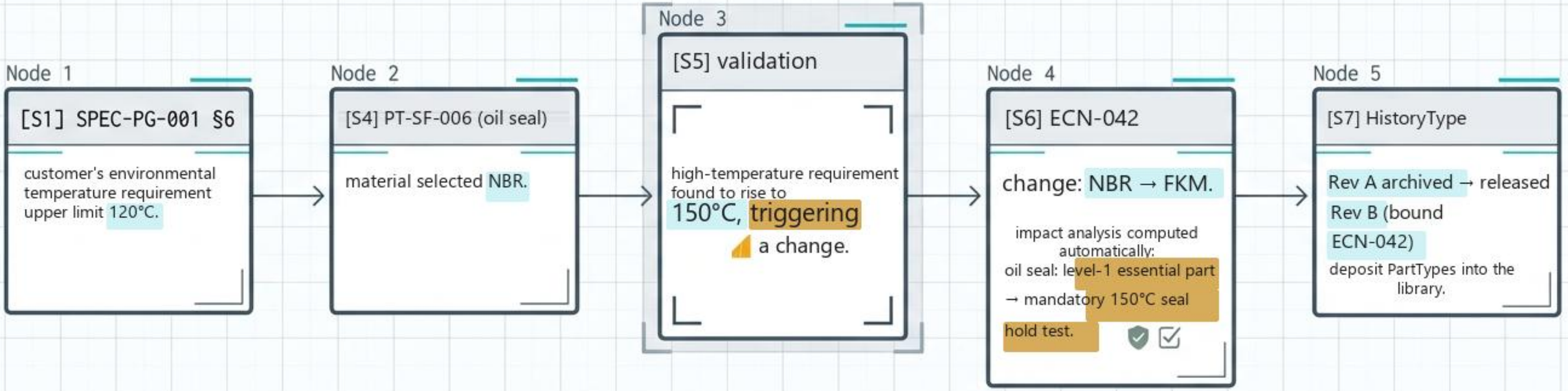
S1-S7 prompt pipeline at a glance

Key rule: each stage does one thing, produces one Type, answers one criterion set.



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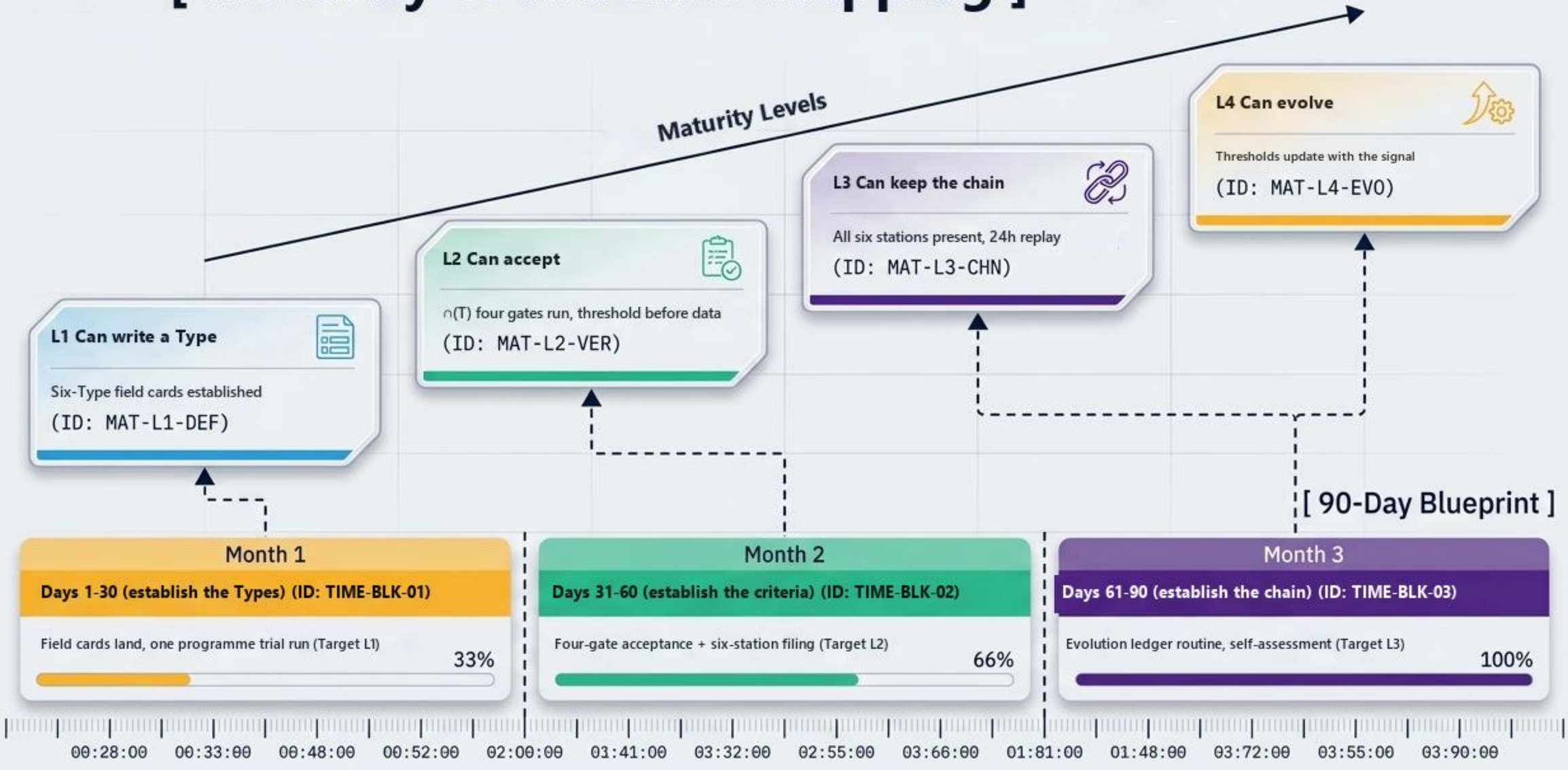
full output synthesis: planetary gearbox S1-S7



Highlight: impact analysis is computed from the dependency chain and the three levels of remove, not memory; revisions stay unbroken (id(C)).

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[Maturity & Timeline Mapping]



[System Status: OPTIMAL | Data Stream: ACTIVE | Last Update: 14:02:35]

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final audit: ten case-closing questions and three wrap-up questions

INSPECTOR'S CLIPBOARD: FINAL AUDIT MATRIX

- ☐ 1. Does every part have a PartType? (list every number)
- ☐ 2. BOM criterion applied? (each item tagged remove level)
- ☐ 3. Container boundaries clear? (housing/module each has id(C))
- ☐ 4. Is every spec quantified? (n(T) clauses measurable)
- ☐ 5. Can every number state its source? ([Source] no blanks)
- ☐ 6. Is the derivation complete? (known→formula→substitution→result, every step checkable)
- ☐ 7. Engineering validation done? (tolerance chain, strength, performance each reported)
- ☐ 6. Is the derivation complete? (known→formula→substitution→result, step checkable)
- ☐ 7. Engineering validation done? (tolerance chain, strength, performance each reported)
- ☐ 8. Does every change carry impact analysis? (ECN has dependency list)
- ☐ 9. Can the history answer "why"? (any revision's release basis)
- ☐ 10. Has the Type Library accumulated? (fed back into the library)

the three wrap-up questions (The Final Synthesis)

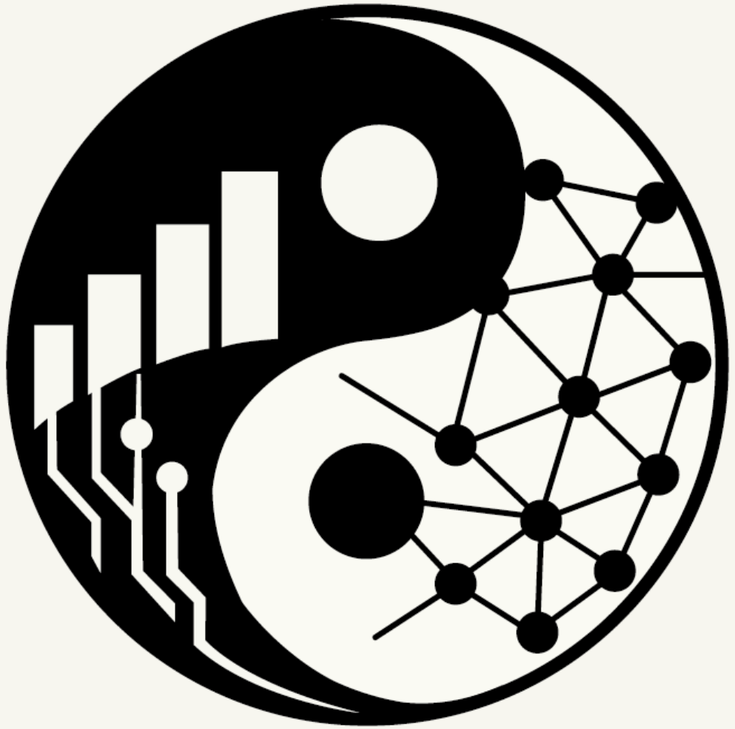
One — are the Types established? Type first, then numbers.

Two — is accuracy held? Four gates of insurance; where the gap is must be clear.

Three — will tomorrow be more accurate? Archiving and evolution are not magic but the drip of each closed case. The next machine must beat this one.

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LonGreat Vision: to create intelligent life that understands the essence of the world



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"We aim to make artificial intelligence not merely a tool, but a system of intelligent life that truly possesses the abilities to understand, reason, learn, and create."

From the unification of knowledge, to the unification of thought, and finally toward the unification of intelligence – this is the eternal path to AGI that LonGreat pursues.

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